

Shihao Weng

PH.D. STUDENT · NANJING UNIVERSITY

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I work on making **LLM agents trustworthy and reliable**, along two directions: **self-evolving agents**, which improve themselves toward more trustworthy behavior over time, and **agent defense**, which protects agents against adversarial inputs and compromised tools.

EDUCATION

Nanjing University 2024 – present

Ph.D. in Software Engineering · Advisor: [Prof. Yang Feng](#)

Nanjing University 2022 – 2024

M.S. in Software Engineering

Jiangnan University 2018 – 2022

B.E. in Computer Science and Technology

RESEARCH VISITS

The Chinese University of Hong Kong Oct 2026 – present

Visiting Ph.D. Student · Host: [Prof. Michael R. Lyu](#)

Singapore Management University Sep 2025 – Sep 2026

Visiting Ph.D. Student · Host: [Prof. Xiaofei Xie](#)

PUBLICATIONS

* corresponding author CCF: China Computer Federation venue rank

Conference Papers

C1 **Are LLM Safety Judges Policy-Invariant? A Three-Principle Stress-Test**

Shihao Weng, Yang Feng*, Xiaofei Xie

Conference on Neural Information Processing Systems (NeurIPS 2026), Evaluations & Datasets Track CCF-A
Spotlight

C2 **TACO: Trust Assessment of Large Language Models in Coding Assistance Tasks**

Shihao Weng, Yang Feng*, Jincheng Li, Yining Yin, Zhenlun Zhang, Lyuxi Liu, Jia Liu

IEEE/ACM International Conference on Software Engineering (ICSE 2026) CCF-A [tool](#)

C3 **AtPatch: Debugging Transformers via Hot-Fixing Over-Attention**

Shihao Weng, Yang Feng*, Jincheng Li, Yining Yin, Xiaofei Xie, Jia Liu

IEEE/ACM International Conference on Software Engineering (ICSE 2026) CCF-A [paper](#) [code](#)

C4 **VADA: A Multicultural Benchmark for Value-Aware Data Generation and Alignment Evaluation in LLMs**

Zhenlun Zhang, Yang Feng*, **Shihao Weng**, Yining Yin, Jincheng Li, Jia Liu

IEEE/ACM International Conference on Software Engineering (ICSE 2026) CCF-A

C5 **Lightweight Probabilistic Coverage Metrics for Efficient Testing of Deep Neural Networks**

Yining Yin, Yang Feng*, **Shihao Weng**, Xinyu Gao, Jia Liu, Zhihong Zhao

Asia-Pacific Symposium on Internetware (Internetware 2025) CCF-C [paper](#) [code](#)

C6 **Dactive: Data Fault Localization for Object Detection Systems**
Yining Yin, Yang Feng*, **Shihao Weng**, Yuan Yao, Jia Liu, Zhihong Zhao
ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2024) CCF-A [paper](#) [code](#)

C7 **Dynamic Data Fault Localization for Deep Neural Networks**
Yining Yin, Yang Feng*, **Shihao Weng**, Zixi Liu, Yuan Yao, Yichi Zhang, Zhihong Zhao, Zhenyu Chen
ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE 2023) CCF-A [paper](#) [code](#)

C8 **Prioritizing Testing Instances to Enhance the Robustness of Object Detection Systems**
Shihao Weng, Yang Feng*, Yining Yin, Jia Liu
Asia-Pacific Symposium on Internetware (Internetware 2023) CCF-C **Distinguished Paper Award**
[paper](#) [code](#)

Journal Articles

J1 **Data Preparation and Quality for Code-Centric Generative Software Engineering Tasks: A Systematic Literature Review**
Shihao Weng, Yang Feng*, Yining Yin, Zhenlun Zhang, Baowen Xu
Frontiers of Computer Science, 2025 CCF-B [paper](#)

J2 **Seeing the Invisible: Test Prioritization for Object Detection System**
Shihao Weng, Yang Feng*, Yining Yin, Yuxuan Dai, Jia Liu, Zhihong Zhao
Empirical Software Engineering 29(6):154, 2024 CCF-B [paper](#) [code](#)

Preprints

P1 **AgentAntibody: An Adaptive Immune System for Defending LLM Agents against Prompt Injection**
Shihao Weng, Yang Feng*, Xiaofei Xie, Jiongchi Yu
arXiv:2608.04053, 2026 [paper](#)

P2 **ARGUS: Defending LLM Agents Against Context-Aware Prompt Injection**
Shihao Weng, Yang Feng*, Jinrui Zhang, Xiaofei Xie, Jiongchi Yu, Jia Liu
arXiv:2605.03378, 2026 [paper](#) [code](#)

HONORS AND AWARDS

Outstanding Graduate Student, Nanjing University (three consecutive years)	2023 – 2025
Distinguished Paper Award, Internetware 2023	2023
National Scholarship for Graduate Students, Ministry of Education of China	2023
CCF Outstanding Undergraduate Student, China Computer Federation	2021
Bronze Medal, 45th ICPC Asia Regional Contest	2021
National Scholarship for Undergraduate Students (two consecutive years)	2020 – 2021

ACADEMIC SERVICE

Reviewer, NeurIPS 2026 Workshop (TAE)	2026
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